

7. Write a program to implement Logistic Regression (LR) algorithm in python

Aim

To implement the **Logistic Regression (LR)** algorithm in Python and evaluate its performance using an appropriate dataset.

Algorithm

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Standardize the input features.
5. Create a Logistic Regression model.
6. Train the model using the training dataset.
7. Predict the class labels for the test dataset.
8. Calculate the accuracy of the model.
9. Display the predicted values and accuracy.

PROGRAM:

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
data = load_breast_cancer()
```

```
X = data.data
```

```
y = data.target
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.3, random_state=42  
)
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Actual Values:")
```

```
print(y_test)
```

```
print("\nPredicted Values:")
```

```
print(y_pred)
```

```
print("\nAccuracy:", accuracy)
```

OUTPUT:

```
print("\nAccuracy:", accuracy)

--- Actual Values:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 0 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0 1 0 1 1 1 0 1 1 1 0 1 0 0 1 1 0 0 0 1 1 1 0 1 1 1 0 1 0 1 1 0 1 0 0
 0 1 0 1 1 1 1 0 0 1 1 1 1 1 1 1 0 1 1 1 1 0 1]

Predicted Values:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 0 0 0 0 1 1 1 0 1 1 1 0 1 0 0 1 1 0 0 0 1 1 1 0 1 1 1 0 1 0 1 1 0 1 0 1 0 0
 0 1 0 1 1 1 1 0 0 1 1 1 1 1 1 1 0 1 1 1 1 0 1]

Accuracy: 0.9824561483588771
```

RESULT:

The **Logistic Regression (LR)** algorithm was successfully implemented in Python. The model was trained and tested using the Breast Cancer dataset and achieved **high classification accuracy (approximately 97%–99%)**. Hence, the Logistic Regression algorithm successfully classified the test samples and demonstrated effective performance on the given dataset.